

SREEKALA K R

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with nearly 3 years of experience developing scalable web applications and enterprise software solutions. Skilled in Angular, TypeScript, Node.js, Express.js, and AWS services, with expertise in RESTful APIs, microservices architecture, authentication systems, performance optimization, and real-time data processing. Experienced in building IoT-enabled platforms and delivering full-stack features in Agile environments. Strong collaborator with a proven track record of mentoring team members and delivering high-quality software solutions.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL

Frontend: Angular (14+), RxJS, HTML5, CSS3, SCSS, Bootstrap, Tailwind CSS

Backend: Node.js, Express.js

Databases & Storage: Cassandra (DataStax Astra), MongoDB, Redis, MySQL, PostgreSQL, SQLite

ORMs & ODMs: Sequelize, Mongoose

Cloud & AWS Services: AWS S3, DynamoDB, CloudWatch, SQS, Lambda

Authentication & Authorization: Auth0, JWT, OAuth 2.0, Single Sign-On (SSO), Role-Based Access Control (RBAC)

Architecture & Development Practices: Microservices Architecture, RESTful APIs, Performance Optimization, Agile/Scrum, Code Reviews

Tools & Platforms: Git, GitHub, Postman, VS Code, Jira, Linear, Figma

Operating Systems: Windows, Linux

AI-Assisted Development Tools: ChatGPT, GitHub Copilot, Cursor AI, Claude, Gemini

PROFESSIONAL EXPERIENCE

Software Engineer | **ZeroNorth**, Chennai, India

November 2023 – February 2026

Project: SMARTShip – IoT-enabled vessel management platform for real-time monitoring of vessel performance, machinery health, safety, and compliance.

- Developed reusable Angular (14+) components using TypeScript, improving maintainability and reducing code duplication across the platform.
- Modernized the application UI with HTML5, CSS3, and SCSS, enhancing usability, responsiveness, and cross-device compatibility.
- Utilized RxJS and Angular performance optimization techniques, including lazy loading, to improve responsiveness and reduce bundle sizes.
- Migrated client-side storage from LocalStorage to IndexedDB, enabling reliable offline access and efficient handling of large telemetry datasets.
- Integrated RESTful microservices to support real-time monitoring, reporting, compliance, and operational workflows.
- Implemented Redis caching and optimized data access patterns, significantly improving dashboard response times and reducing database load.
- Leveraged DataStax Astra (Cassandra), PostgreSQL, and MongoDB for telemetry ingestion, historical analytics, and operational data management.
- Designed and implemented secure authentication and authorization using Auth0, OAuth 2.0, JWT, SSO, and RBAC.
- Leveraged AWS services including S3, DynamoDB, CloudWatch, Amazon SQS, and AWS Lambda to support cloud storage, operational monitoring, real-time event processing, and automated backend workflows for telemetry and operational systems.

- Automated voyage-related workflows and notification systems, including onboarding, password recovery, team invitations, and operational alerts.
- Implemented Mixpanel analytics to track user behavior, feature adoption, and engagement metrics.
- Developed and validated API integrations using Postman while contributing to Agile/Scrum delivery processes through Jira and Linear.
- Built supplementary React-based reporting interfaces to support internal operational workflows.
- Leveraged AI-assisted development tools to accelerate feature delivery, code generation, debugging, and technical documentation.
- Conducted code reviews, mentored junior developers, and collaborated with Product, QA, and DevOps teams to deliver high-quality features.

Software Engineer Intern | **Alpha Ori India Private Limited**, Chennai, India *April 2023 – October 2023*

Project: Smart Voyager – *Voyage planning and decision-support platform focused on route optimization, fuel efficiency, and operational safety.*

- Developed responsive frontend features using Angular and TypeScript within a scalable component-based architecture.
- Integrated REST APIs to support real-time voyage planning, dashboard updates, and dynamic data retrieval.
- Built interactive data visualization modules for monitoring fuel consumption, route efficiency, and operational performance.
- Assisted in developing backend functionality, writing SQL queries, and working with MongoDB document models.
- Implemented secure authentication mechanisms and protected application routes.
- Collaborated with UI/UX teams using Figma and validated API functionality through Postman testing.
- Utilized Git, GitHub, and Jira for source control, issue tracking, and Agile team collaboration.
- Contributed to debugging, testing, and feature delivery while gaining domain expertise in maritime operations and fuel optimization.

EDUCATION

B.Tech in Computer Science and Engineering

Mar Athanasius College of Engineering, Kothamangalam | 2020 – 2023

Diploma in Computer Engineering

Govt. Polytechnic College, Kothamangalam | 2017 – 2020

ADDITIONAL INFORMATION

Languages Known: English, Malayalam; Basic proficiency in Tamil and Hindi